

Thibault Pautrel

Postdoctoral researcher in applied mathematics and computer science
Riemannian optimization and federated learning

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Born 17 August 1994 in Le Mans, France — French citizen

PROFESSIONAL EXPERIENCE

2025–present	Postdoctoral researcher CentraleSupélec, Laboratory of Signals and Systems (L2S) — Gif-sur-Yvette <i>Project:</i> Federated learning on covariance matrices for pre-trained models on health data. <i>Supervision:</i> Florent Bouchard.
2025 6 months	Research internship (MSc) LISTIC, Université Savoie Mont Blanc — Annecy <i>Project:</i> Riemannian Federated Learning on remote sensing data. <i>Supervision:</i> Guillaume Ginolhac, Ammar Mian, Florent Bouchard.
2021–2024 Full time	Mathematics teacher (agrégé) French national education system Teaching in secondary schools. Weekly oral examinations in undergraduate preparatory classes (BCPST, PCSI, MP).
2018–2021	PhD in Mathematics Université de Rennes 1, IRMAR <i>Title:</i> Some contributions to the study of the universality of the zeros of random trigonometric functions (defended 10 January 2022). <i>Advisors:</i> Jürgen Angst and Guillaume Poly. <i>Funding:</i> Doctoral contract with a teaching assignment (64 h/year).
2018 5 months	Research internship (MSc) LAMA, Université Savoie Mont Blanc — Chambéry <i>Project:</i> Mean-reflected backward stochastic differential equations. <i>Supervision:</i> Philippe Briand.

PUBLICATIONS

Computer science (artificial intelligence & geometric learning)

Keywords	Federated learning, Riemannian manifolds, SPD matrices, geometric deep learning, optimization on manifolds, signal processing.
[C2] Accepted	T. Pautrel , F. Portier. <i>Riemannian stochastic optimization for sufficient dimension reduction</i> . ICML 2026
[C1] Accepted	T. Pautrel , F. Bouchard, A. Mian, G. Ginolhac. <i>FedSPDnet: geometry-aware federated deep learning with SPDnet</i> . EuSIPCO 2026.

Mathematics (stochastic processes & asymptotic analysis)

Keywords	Nodal sets, random trigonometric polynomials, stationary Gaussian processes, strong dependence, universality, Kac–Rice formulas.
[R2] To appear	J. Angst, T. Pautrel , G. Poly. <i>Global universality of the expected number of zeros of non-analytic random signals</i> . INdAM–Springer Proceedings, 2025.

[R1] Published	J. Angst, T. Pautrel , G. Poly. <i>Real zeros of random trigonometric polynomials with dependent coefficients</i> . Transactions of the American Mathematical Society 375 (2022), 7209–7260.
[R0] Published	T. Pautrel . <i>New asymptotics for the mean number of zeros of random trigonometric polynomials with strongly dependent Gaussian coefficients</i> . Electronic Communications in Probability 25 (2020), no. 36.

TEACHING

University teaching

2026–2027 6 h TD-eq.	Labs: Machine Learning for Data Science ENSAI — MSc Smart Data 17 students.
2025–2027 33 h TD-eq.	Labs: Introduction to C++ programming IUT Orsay — 1st year (R101-B) 15 students, 5.5 h per week.
2025–2026 12 h TD-eq.	Tutorials and labs: Machine Learning CentraleSupélec — 2nd year 102 students, taught in English, 8 sessions of 1.5 h.
2025–2026	Lectures and tutorials: Statistics and Learning CentraleSupélec — 1st year ≈30 students, taught in French and English (S2). Lectures: $2 \times 1.5 \text{ h} = 4.5 \text{ h}$ TD-eq. + 15 h of tutorials.
2025–2026 1.5 h TD-eq.	Tutorials: Digital image processing CentraleSupélec — 2nd year Taught in French.
2018–2021 48 h TD-eq.	Tutorials: Analysis and probability Université de Rennes 1 — 2nd year, Mathematics ≈30 students, 4 h per week.
2018–2021 16 h TD-eq.	Tutorials: Probability Université de Rennes 1 — 1st year, Biology ≈30 students, 2 h per week over 8 weeks.

Supervision and examination boards

2026	Master's thesis examination board ENSAI — MSc Smart Data
2023–2025	Examiner, CentraleSupélec entrance examination Marker for the written MP Mathematics 1 papers (2023, 2024, 2025). Examiner for the Mathematics–Computer Science oral examinations (2023, 2024).
2019–2020	Supervision of undergraduate research projects ENS Rennes — 1st year Topic: Stein's method for normal approximation.
2018–2021	Mock oral examination board Université de Rennes 1 Preparation for the CAPES and the <i>agrégation</i> in mathematics.

Oral examinations in undergraduate preparatory classes

2021–2024	Weekly oral examinations in mathematics BCPST, PCSI, MP — various institutions
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2016–2018 **Weekly oral examinations in mathematics**
MP, MP* — Lycée Chateaubriand, Rennes

Secondary school teaching

2021–2024 **Mathematics teacher (*agrégé*)**
Full time Secondary schools

EDUCATION

2024–2025 **MSc Smart Data**
ENSAI, Rennes — with highest honours
Specialisation: machine learning, deep learning, dimension reduction, functional data analysis.

2017–2018 **MSc in Probability and Statistics (“Aléatoire”)**
Université de Rennes 1 — with honours
Specialisation: stochastic processes and stochastic calculus, statistics of processes, dynamical systems, parametric and non-parametric statistics, Malliavin calculus and Stein’s method

2016–2017 **Preparation for the *agrégation* in mathematics**
Université de Rennes 1 — with honours

2015–2016 **MSc (first year) in pure mathematics**
Université de Rennes 1 — with honours

2014–2015 **BSc in mathematics**
Université de Rennes 1 — with honours

2012–2014 **Undergraduate preparatory classes, MPSI – MP***
Lycée Chateaubriand, Rennes

COMPETITIVE EXAMINATIONS AND DISTINCTIONS

2026 **National qualification for Associate Professor positions**
CNU section 26 — applied mathematics and applications of mathematics

2017 ***Agrégation externe* in mathematics**
Probability–statistics option — rank: 118

SERVICE

2026 **Reviewer**
ICML 2026 — International Conference on Machine Learning

2019–2021 **Co-organiser of the doctoral seminar**
“Gaussbusters” seminar — IRMAR, Rennes
Weekly seminar of the PhD students in probability and statistics.

TECHNICAL SKILLS

Languages Python (NumPy, PyTorch, scikit-learn), R, SQL, C++, OCaml, Scilab

Distributed Spark, parallel computing, HTCondor scheduler
computing
Tools Git, $\text{\LaTeX}$, Unix/Bash

LANGUAGES

French Native

English Fluent (C2) Certificate of Proficiency in English (Cambridge, 2013)

Spanish Intermediate (B2) Read, written, spoken